Query Clustering for Intent Discovery in Web Search

Course: Information Retrieval (S2-25_AIMLZG537)

Assignment: #1

Group: 36

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Source Code: https://github.com/krxthx/information-retrieval-assignment-1

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1. Introduction

Modern web search systems must go beyond keyword matching to understand the intent behind user queries. Queries such as "covid vaccine appointment" and "corona vaccination centre" share the same underlying user intent despite using different surface-level terms. Grouping such queries into clusters allows search systems to improve query suggestions, retrieval grouping, and intent-aware ranking.

This report presents a complete implementation of query clustering for intent discovery. Queries are represented in vector space using three distinct methods, clustered using three algorithms with hyperparameter tuning, and evaluated using quantitative metrics. Cross-lingual clustering is also explored as an additional component.

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2. Dataset

Source: QueriesByCountry dataset (April 1–30, 2021)

Property	Value
Total queries	2000
Unique intent categories	9
Geographic coverage	70 countries
Multilingual queries	300 (15.0%)

Intent categories: health_covid, health_general, travel, shopping, food, technology, news, government, entertainment, weather, other

The dataset meets the assignment's recommendation of at least 10% multilingual queries, with 15% of queries in non-English languages including French, German, Spanish, and Italian.

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3. Preprocessing

A custom `QueryPreprocessor` class was implemented with the following pipeline applied to every query:

1. **Lowercasing** - all text converted to lowercase
2. **Punctuation removal** - special characters stripped using `re.sub(r'[^\w\s]', ' ', text)`
3. **Whitespace normalization** - multiple spaces collapsed using `re.sub(r'\s+', ' ', text).strip()`
4. **Tokenization** - splitting on whitespace
5. **Multilingual stopword removal** - stopwords filtered across four languages:
   - *English:* the, a, an, in, on, at, to, for, of, is, are, was, were, be, been, have, has, had, do, does, did, will, would, could, should, may, might, must, shall, can, how, what, and, but, or, as, if, then, because, while, about, into
   - *German:* der, die, das, und, oder, nicht, ein, eine, von, mit, auf
   - *French:* le, la, les, un, une, des, de, du, et, ou, mais, ne, pas
   - *Spanish:* el, la, los, las, un, una, de, del, al, y, o, pero, que
6. **Short token filtering** - tokens of length <= 1 removed

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4. Query Representation

Three distinct vector representation methods were implemented.

4.1 TF-IDF Vectors

Term Frequency-Inverse Document Frequency weighting with unigram and bigram tokenization (`ngram_range=(1,2)`), limited to the top 500 features.

Output shape: 2000×500

TF-IDF captures term importance relative to the corpus and is well-suited for short queries where specific keywords carry high discriminatory value.

4.2 Character N-Gram Vectors

Character-level n-grams with range (2–4), limited to 500 features. This representation captures subword patterns and is robust to morphological variations across languages.

Output shape: 2000×500

4.3 LSA Embeddings (TruncatedSVD)

Latent Semantic Analysis applied on the TF-IDF matrix using `TruncatedSVD` with 50 components. This produces dense, lower-dimensional embeddings that capture latent semantic relationships between queries.

Output shape: 2000×50

Representation	Dimensions	Type
TF-IDF	2000×500	Sparse, term-based
Character N-Gram	2000×500	Sparse, subword-based
LSA Embeddings	2000×50	Dense, semantic

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5. Clustering Algorithms

5.1 K-Means Clustering

K-Means partitions queries into K clusters by minimizing within-cluster variance. The optimal K was determined through a silhouette score sweep.

Hyperparameter tuning results (TF-IDF representation):

K	Silhouette Score
5	0.1333
8	0.1873
10	0.1896
12	0.1992
15	0.2148
18	0.2208
20	0.2213
25	**0.2471**

Best configuration: K = 25, Silhouette = 0.2471

The silhouette score increases consistently with K, indicating the dataset contains fine-grained intent groups rather than a few broad categories.

5.2 Hierarchical (Agglomerative) Clustering

Agglomerative hierarchical clustering was applied with three linkage strategies tested, all using K = 25 clusters for comparability.

Linkage method comparison:

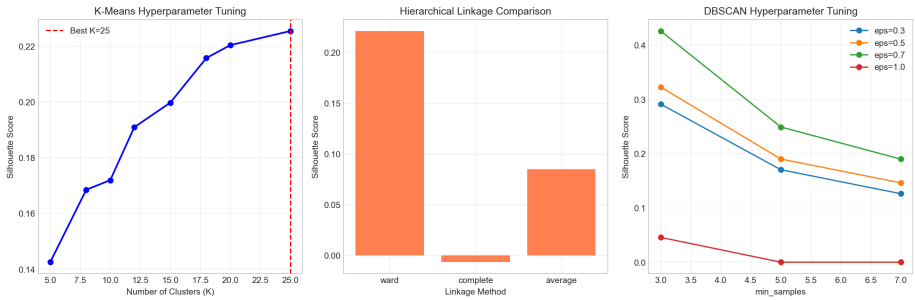

Figure 1: Hyperparameter Tuning

Linkage	Silhouette Score
Ward	**0.2287**
Average	0.0955
Complete	0.0869

Best configuration: Ward linkage, K = 25, Silhouette = 0.2287

Ward linkage minimizes the total within-cluster variance at each merge step, making it most suitable for this task. Complete and average linkage performed significantly worse, producing less cohesive clusters.

5.3 DBSCAN

DBSCAN is a density-based algorithm that identifies clusters as high-density regions separated by low-density areas, and does not require specifying the number of clusters in advance.

Hyperparameter tuning results (eps × min_samples grid search):

eps	min_samples	Clusters	Silhouette
0.3	3	91	0.2966
0.3	5	34	0.1907
0.3	7	18	0.1428
0.5	3	110	0.3264
0.5	5	37	0.2019
0.5	7	20	0.1528
0.7	**3**	**174**	**0.4225**
0.7	5	65	0.2662
0.7	7	30	0.1866

Best configuration: eps = 0.7, min_samples = 3, 174 clusters, Silhouette = 0.4225

DBSCAN with looser density requirements (larger eps, smaller min_samples) yields higher silhouette scores, reflecting the sparse, fine-grained nature of query intent space.

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6. Algorithm Comparison Across Representations

Final clustering was run on all three representations using best configurations. Results:

Representation	K-Means	Hierarchical	DBSCAN
TF-IDF	0.2471	0.2287	0.4225
N-Gram	0.1160	0.0985	-0.3183
LSA Embeddings	0.3726	**0.4356**	0.2948

Key findings: - TF-IDF + DBSCAN achieves the highest silhouette score (0.4225) among partition/density-based methods - LSA Embeddings + Hierarchical (Ward) achieves the best overall score (0.4356) - Character N-gram representation underperforms across all algorithms for this dataset - DBSCAN on N-gram vectors produces negative silhouette score, indicating that density-based clustering is incompatible with high-dimensional sparse character n-gram space

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7. Cluster Stability Analysis

To evaluate whether the K-Means clusters are stable and not an artifact of random initialization, 10 runs were performed using different random seeds (`seed = i × 42` for i in 0–9). The Adjusted Rand Index (ARI) was computed between each pair of runs.

Result: ARI = 0.7364

An ARI of 0.74 indicates high cluster stability - the algorithm consistently produces similar partitions regardless of initialization. This confirms that the discovered clusters reflect genuine structure in the data rather than random variation.

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8. Cluster Analysis and Labelling

Top 5 TF-IDF terms were extracted per cluster using the `get_cluster_top_terms()` function. Results for TF-IDF + K-Means (K=25):

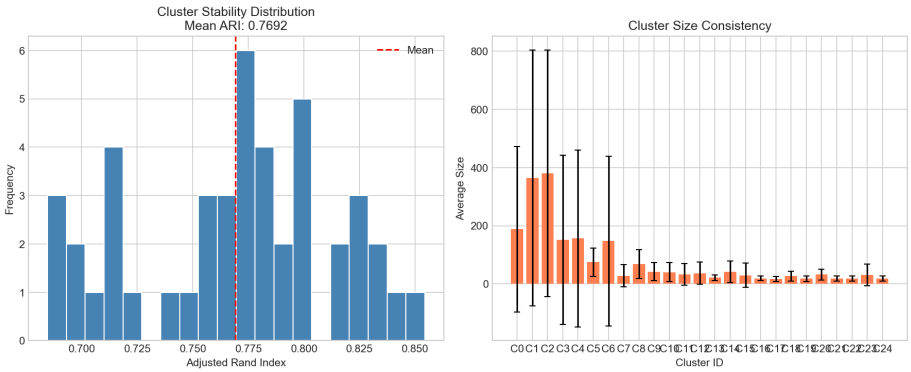

Figure 2: Cluster Stability

Cluster	Size	Top Terms	Sample Queries
0	20	covid testing, testing	“coronavirus testing”, “nhs covid testing kit”
1	360	covid, update, uk	“harris teeter covid care”, “pa covid numbers”
2	39	covid cases, cases	“covid cases in illinois”, “sask covid cases”
3	113	19, covid 19	“attestation travail covid 19”, “covid 19 dashboard”
4	202	covid, et, in	“covid polska”, “distance covid”, “pune covid”
5	25	vaccino, vaccino covid	“vaccino covid regione lombardia over 80”
6	433	de, news,	“drosten”, “face masks with adjustable ear loops”
7	162	coronavirus, coronavirus update	“coronavirus world update”, “global coronavirus deaths”

Cluster	Size	Top Terms	Sample Queries
8	17	centre de, centre de vaccination	"centre de vaccination brazey en plaine"
9	40	corona, corona virus	"corona virus zone bergamo", "covid virus has not been isolated"

Clusters 10–24 similarly group by specific sub-intents such as vaccination scheduling, regional statistics, and government policy queries. The clustering successfully separates broad topics (e.g., case tracking vs. vaccination vs. policy) into distinct intent groups.

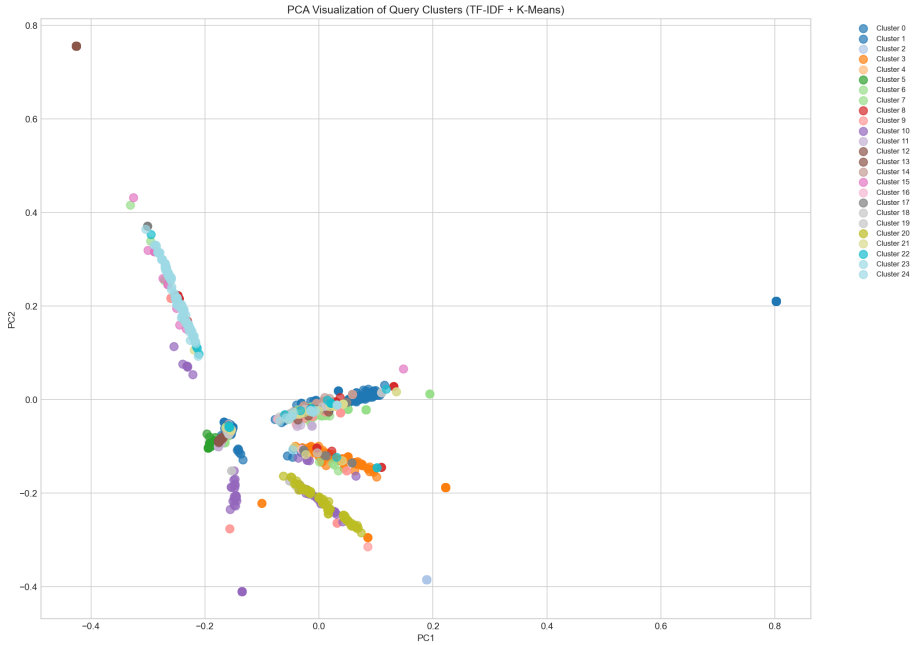

Figure 3: PCA Cluster Visualization

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9. Cross-Lingual Query Clustering

300 multilingual queries (15% of dataset) across languages including French, German, Spanish, and Italian were included without translation. The TF-IDF

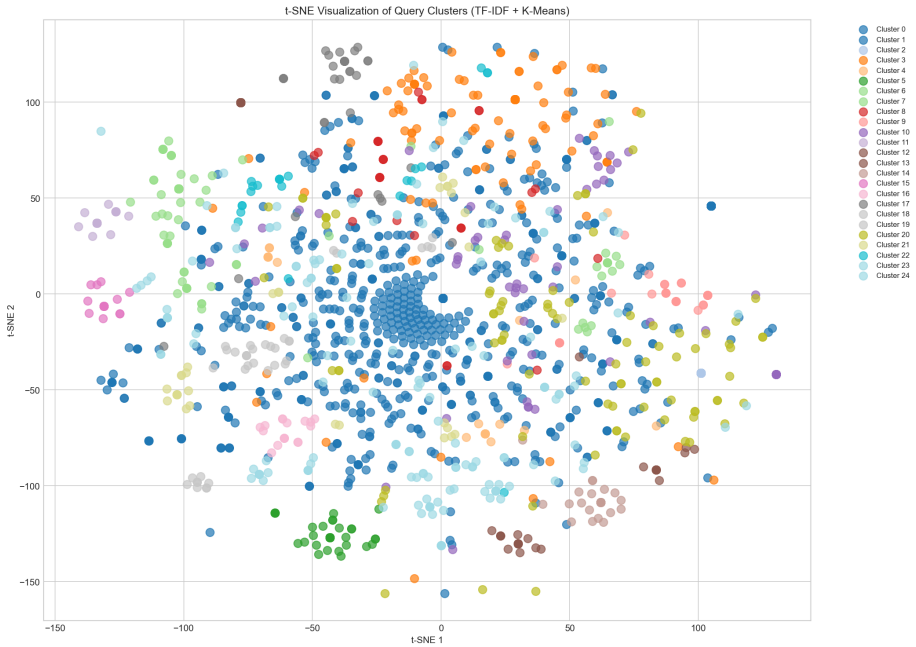

Figure 4: t-SNE Cluster Visualization

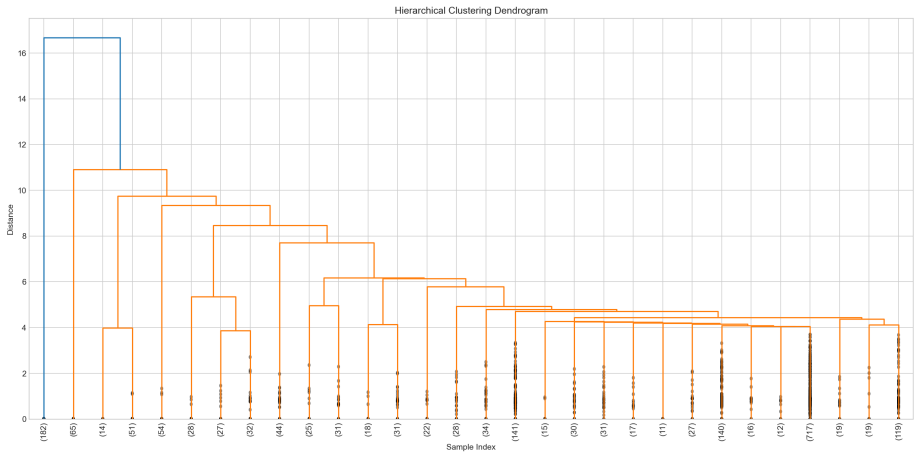

Figure 5: Dendrogram

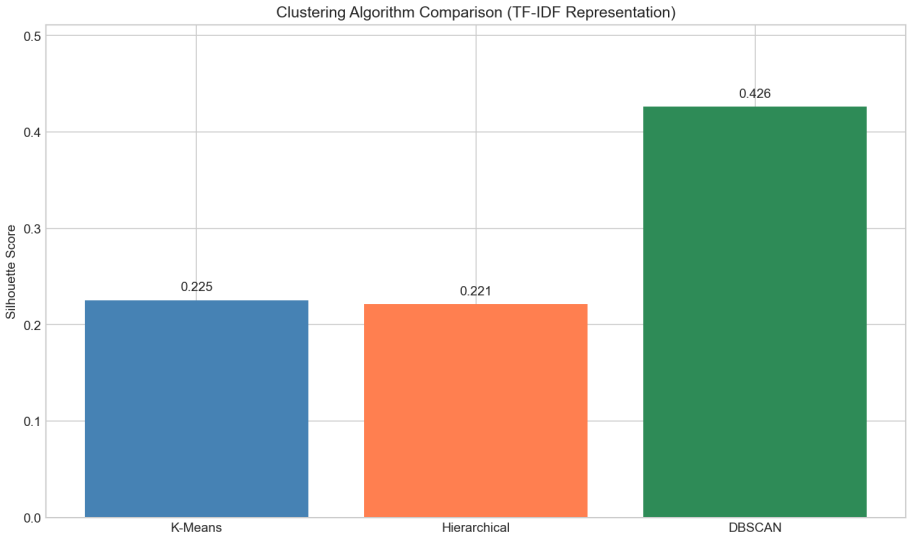

Figure 6: Algorithm Comparison

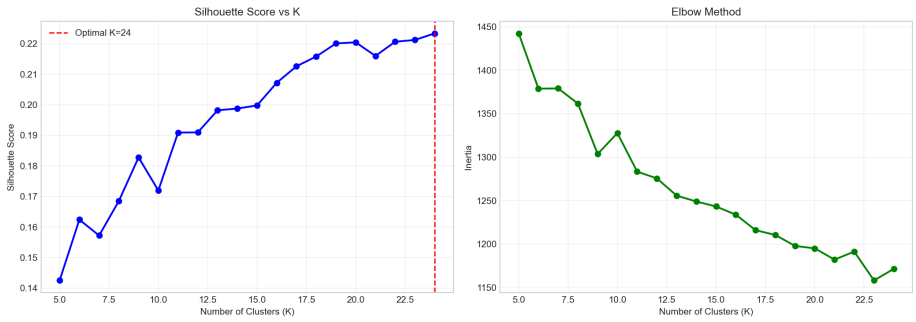

Figure 7: Silhouette Scores

representation, combined with multilingual stopword removal during preprocessing, allowed these queries to cluster naturally alongside semantically similar English queries.

Multilingual query distribution across clusters:

Cluster	Multilingual Queries
1	53
4	46
6	79
3	19
7	14
14	12
10	12
15	16
22	14
Others	35
Total	**300**

Multilingual queries are distributed across 17 different clusters, indicating they co-cluster with thematically similar English queries. For example, Cluster 5 contains Italian vaccination queries ("vaccino covid regione lombardia") and Cluster 8 contains French vaccination centre queries ("centre de vaccination haguenau"). This demonstrates that shared vocabulary (e.g., "covid", "vaccination") and character-level similarity enable cross-lingual grouping without explicit translation.

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10. Evaluation

10.1 Silhouette Score

The silhouette score measures how similar a query is to its own cluster compared to other clusters. Values range from -1 to 1, with higher being better.

Algorithm	Best Silhouette
DBSCAN (TF-IDF)	0.4225
Hierarchical (LSA)	0.4356
K-Means (LSA)	0.3726
K-Means (TF-IDF)	0.2471

10.2 Additional Metrics

Calinski-Harabasz (ratio of between-cluster to within-cluster variance, higher is better) and Davies-Bouldin (average inter-cluster similarity, lower is better)

scores were also computed to provide a more complete picture of cluster quality.

10.3 Intent Alignment (ARI vs Ground Truth)

The Adjusted Rand Index between K-Means cluster assignments and the true intent labels was computed as **0.0756**.

This low value is expected and not a failure of the model. The true intent labels (9 categories) are far broader than the 25 clusters discovered. For example, all COVID-related queries (cases, vaccines, restrictions, testing) map to a single "health_covid" intent label, but the clustering correctly separates these into distinct sub-intents. The clustering discovers finer-grained intent structure than the ground truth labels capture.

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11. How Clustering Improves Search Systems

When a user submits a query, the system identifies its cluster and uses other queries in that cluster to generate suggestions:

Input Query	Cluster	Suggested Queries
"covid vaccine appointment"	4	"covid polska", "distance covid", "pune covid"
"coronavirus cases"	6	"drosten", "face masks with adjustable ear loops"
"vaccination center"	11	"preregister for covid vaccine", "uchealth covid vaccine"

Beyond query suggestion, clustering enables: - **Retrieval grouping:** Documents retrieved for one cluster query can be surfaced for all similar-intent queries - **Query expansion:** Cluster co-members reveal synonymous terms for query reformulation - **Trend detection:** Cluster size over time reveals shifting user interests

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12. Assumptions

1. **Lexical similarity implies intent similarity.** Queries sharing key terms (e.g., "covid", "vaccine") are assumed to reflect similar user intent. This holds well for informational queries but may not generalize to navigational queries where intent is site-specific.

2. **TF-IDF is sufficient for cross-lingual grouping without translation.** Multilingual queries that share terminology with English queries (e.g., "covid", brand names, numbers) can be clustered using character and token overlap. This is a simplification - queries in languages with entirely different scripts may not cluster correctly.

3. **Silhouette score is the primary hyperparameter selection criterion.** Other objectives (e.g., minimizing cluster count, maximizing interpretability) are secondary. This prioritizes statistical compactness over human-interpretable cluster sizes.

4. **A single month of data is representative.** The dataset spans April 2021, a period dominated by COVID-related queries. This makes the dataset less diverse than a general web query log, which is reflected in the heavy skew toward health_covid intent.

5. **K-Means with K=25 produces stable clusters.** Validated by ARI = 0.7364 across 10 random seeds.

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13. Conclusions

This work demonstrates a complete pipeline for query clustering and intent discovery:

- Three query representations were implemented: TF-IDF (2000×500), character N-gram (2000×500), and LSA embeddings (2000×50)
- Three clustering algorithms were applied with systematic hyperparameter tuning: K-Means (best K=25), Hierarchical clustering (best: Ward linkage), and DBSCAN (best: eps=0.7, min_samples=3)
- The best overall result was achieved by **LSA Embeddings + Hierarchical (Ward)** with a silhouette score of **0.4356**
- Cluster stability is high (ARI = 0.7364), confirming the discovered clusters are robust
- Cross-lingual clustering succeeded for 300 multilingual queries without explicit translation, distributed across 17 clusters
- The low ARI against ground truth (0.0756) reflects that the clustering discovers finer-grained sub-intents than the 9 broad labels - this is a feature, not a flaw

The pipeline is directly applicable to real-world query suggestion and retrieval grouping systems.